
> Interlock

Detailed Business Proposal. Runtime assurance for agentic AI in insurance operations.

Accenture Innovation Challenge 2026 Round 2, Prototype Development ControlPlane.ai

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Contents: problem framing, solution design, target users, business case and impact, phased roadmap, key risks with mitigations.

1 The commercial problem

1.1 The bottleneck is not capability

Insurers are not slow to automate claims because the models are not good enough. They are slow because nobody can prove the automation is safe. Accenture's own Technology Vision 2025 research records that seventy-seven percent of executives believe the benefits of AI are only possible on a foundation of trust, and states the position plainly: systems will only ever be as autonomous as they are trustworthy.

The consequence is a stalled pipeline. Organisations build capable claims agents, then deploy them behind mandatory human review for every decision, which removes most of the economic case for building them. The agent becomes a suggestion engine, and the cost of the human review remains.

THE COMMERCIAL INSIGHT

Governance is currently treated as a brake on autonomy. Interlock reframes it as the throttle. An organisation that can prove a bounded error rate on unattended decisions can safely increase the proportion of decisions taken without a human, which is where the return actually sits.

1.2 Quantified exposure

EXPOSURE	EVIDENCE AND MECHANISM
Claims leakage	Insurers already lose an estimated five to ten percent of claim payouts to leakage, being amounts paid that should not have been paid or should have been lower. An autonomous agent operating at machine speed amplifies this without a corresponding increase in visibility
Regulatory liability	The EU AI Act classifies insurance risk assessment and pricing AI as high risk, with logging obligations under Article 12 and human oversight obligations under Article 14 applying from August 2026. A wrongful automated denial is a regulatory event, not a customer service issue

Established precedent	A Canadian tribunal has held an airline liable for a commitment made by its chatbot. Organisations own what their AI states. An agent that executes converts that liability from an apology into a payout
Foregone efficiency	Every claim routed to a human that did not require one carries a handling cost and a cycle-time penalty. This is the largest and least discussed cost of ungoverned AI

2 Solution design

2.1 What Interlock is

Interlock is a runtime assurance layer that sits between an AI agent and the systems it acts upon. It is deployed as an inline gate, not as a monitoring dashboard. No action reaches a payment rail, a policy database or a customer until Interlock has verified it and assigned it to an autonomy lane.

The architecture follows the Simplex pattern used to certify avionics autopilots, comprising an untrusted controller, a verified safety monitor and a safe fallback. The agent must declare a structured intent contract before acting. Six independent verification checks then execute in parallel, are fused into a four-axis risk vector, and are routed into one of five graduated autonomy lanes ranging from immediate execution to fail-closed refusal.

2.2 What makes it defensible

DIFFERENTIATOR	WHY COMPETITORS DO NOT HAVE IT
Governs actions, not text	The established guardrail market inspects model output for toxicity and accuracy. Interlock inspects the tool call, which is where irreversible consequences occur
Heterogeneous verification	Six checks spanning deterministic rules, a classifier, two language model judges and a statistical uncertainty measure. Correlated detectors fail together, so method diversity is a safety property
Cross-vendor Two-Key	Irreversible actions require concurrence from a model belonging to a different vendor, running on different infrastructure, under a separate credential. A single poisoned model or leaked key cannot authorise an action
Derived thresholds	The business sets the acceptable error rate. The routing threshold is computed from it by split-conformal calibration. No engineer hand-tunes a sensitivity dial
The compounding asset	Every verdict and every human override is a labelled example. The moat is not the architecture, which is publishable. The moat is the verdict ledger, which is proprietary and grows with usage

2.3 Position relative to the Accenture portfolio

Accenture’s Trusted Agent Huddle certifies and scores agents before they join the enterprise, providing interoperability across platforms through the Agent2Agent and Model Context Protocol standards. Interlock is complementary rather than competitive. Huddle establishes which agents are trusted to participate. Interlock governs every action those agents take thereafter. The two together provide admission control and runtime control, which is the pairing regulated deployment requires.

3 Target users

USER	WHAT THEY NEED	WHAT INTERLOCK GIVES THEM
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Claims operations lead Economic buyer	To raise straight-through settlement without raising leakage	A measured straight-through rate against a bounded error rate, with the autonomy dial as a direct control
Claims reviewer Daily user	To clear a queue quickly without missing anything material	Only genuinely uncertain actions arrive, with evidence already prepared and keyboard-speed disposition
Chief risk and compliance officer Approver	Demonstrable Article 12 and Article 14 conformity, and cohort fairness evidence	A tamper-evident ledger, replayable decisions, jurisdiction-specific policy packs and a drift sentinel
AI platform engineer Implementer	To govern agents without rewriting them	An inline gate at the tool-call boundary, model-agnostic and requiring no access to model internals
Internal auditor Periodic user	To answer why a specific decision was taken, months later	Full replay of any action showing the declared plan, every check, the evidence and the resulting ledger entry

4 Business case

BASIS OF PREPARATION

The Round 2 brief states that teams are not expected to have access to proprietary company data and should make reasonable assumptions and state them clearly. The following model is illustrative and built on stated public benchmarks. Unit economics for inference cost, latency and detection accuracy are measured directly from the prototype. Volume and handling-cost assumptions are modelled.

4.1 Reference organisation

ASSUMPTION	VALUE	SOURCE
Annual claims settled	1,560,000	30,000 per week, per the brief
Proportion routed through AI agents by year three	60%	Modelled adoption
Fully loaded cost of one human claim touch	EUR 18	Modelled, European operations
Average settled claim value	EUR 1,900	Modelled, motor and home mix
Baseline leakage rate	6%	Midpoint of the published range
Straight-through rate achieved	54.5%	Measured in the prototype
Inference cost per governed action	\$0.00164	Measured in the prototype

4.2 Value model, steady state at year three

VALUE DRIVER	CALCULATION	ANNUAL VALUE
Avoided human handling	936,000 agent-routed claims, 54.5 percent settled without a human touch, at EUR 18 per avoided touch	EUR 9.2m
Leakage prevented	Governance improves decision accuracy by 9.1 percentage points. Applied conservatively to one quarter of the leakage exposure on agent-routed volume	EUR 6.1m
Regulatory penalty avoidance	Treated as risk reduction rather than as a modelled cash benefit	not modelled
Cycle-time improvement	Faster settlement supports retention. Not modelled, to keep the case conservative	not modelled

4.3 Cost model

COST LINE	BASIS	YEAR 1	YEAR 3
Inference	936,000 actions at measured unit cost, with production model pricing	EUR 0.02m	EUR 0.05m
Platform and infrastructure	Verification service, ledger storage, observability	EUR 0.35m	EUR 0.55m
Engineering and integration	Build, policy authoring, systems integration	EUR 0.90m	EUR 0.60m
Governance operations	Policy stewardship, review of drift and fairness reports	EUR 0.20m	EUR 0.30m
Total cost		EUR 1.47m	EUR 1.50m

Inference is a rounding error in the model. This is the important structural point. The measured cost of governing an action is approximately one sixth of a US cent, which is four orders of magnitude below the cost of the human review it replaces. The economics of runtime assurance are not constrained by compute.

4.4 Return

EUR 15.3m	EUR 1.5m	10.2x	< 6 mo
Gross annual value at steady state	Annual run cost, steady state	Return on cost, year three	Payback from pilot go-live

Sensitivity is dominated by the straight-through rate. If straight-through were to reach only thirty percent rather than the measured 54.5 percent, gross annual value falls to approximately EUR 9.2m and the return on cost remains above six times. The case does not depend on optimistic detection performance.

5 Impact beyond the financial case

DIMENSION	IMPACT
Customer	Legitimate claims settle in minutes rather than days, because governance permits unattended settlement. Wrongful denials fall, since adverse automated decisions are routed to a human under the EU configuration
Workforce	Reviewers receive only genuinely uncertain cases, with evidence prepared. The role shifts from processing to judgment, which is both higher value and more defensible
Regulatory	Conformity becomes demonstrable rather than asserted. Any decision can be replayed, and the audit trail is tamper evident rather than merely access controlled
Fairness	The drift sentinel reports the unattended approval rate by cohort continuously, converting fairness from a periodic audit into an operational metric
Sector	The pattern generalises. Nothing in the architecture is specific to insurance. The same mesh governs a support assistant and an internal copilot in the prototype, driven only by different policy packs

6 Phased roadmap

PHASE	DURATION	OBJECTIVE AND SCOPE	EXIT CRITERION
Phase 0 Prototype	Complete	Demonstrate the core mechanism on a simulated insurer. Six-check mesh, five-lane routing, cross-vendor Two-Key, signed ledger, conformal recalibration, operator console	Achieved. Zero false negatives, 14.3 percent false positive rate
Phase 1 Shadow	3 months	Deploy inline against one live claims agent in observe-only mode. Interlock records a verdict for every action but blocks nothing. Policy packs authored with the compliance function	Verdict agreement with human reviewers above ninety percent
Phase 2 Enforce, low stakes	3 months	Enable enforcement for reversible and low-value actions. Introduce the EDIT lane. Calibrate tau from accumulated labelled verdicts rather than defaults	Straight-through above forty percent with no confirmed false negative
Phase 3 Enforce, full	6 months	Extend to irreversible actions with Two-Key concurrence. Add the mutual TLS gateway and Open Policy Agent compilation. Add a third judge vendor	Straight-through above fifty-five percent, conformity pack accepted by the risk committee
Phase 4 Multi use case	6 months	Onboard the support assistant and internal copilot using their own policy packs. Introduce champion and challenger shadow evaluation and LoRA distillation of judges	Three governed use cases, unit cost reduced by half
Phase 5 Scale	Ongoing	Additional geographies through jurisdiction packs. Integration with Trusted Agent Huddle so certified agents inherit a governance profile on admission	Governance provisioned as a platform capability

7 Key risks and mitigations

RISK	SEVERITY	MITIGATION
Alert fatigue Over-escalation trains staff to bypass the system	High	Measured and managed as a first-class metric. Our own first evaluation produced a 41.2 percent false positive rate, which two-key conservative reconciliation and the EDIT lane reduced to 14.3 percent with no change in false negatives. The Trust Report reports this continuously, and the acceptable error rate is set by the business rather than by engineering
Correlated judge failure Both keys share a blind spot	High	Method diversity across deterministic rules, a classifier, language model judges and a statistical measure. Vendor diversity on the second key, on separate infrastructure and a separate credential. A third vendor is scheduled in Phase 3. Irreversible actions additionally fail closed rather than defaulting to execution
Latency budget breach	Medium	Checks execute in parallel, and the measured serial-to-parallel saving is reported per action. The deterministic fast path completes within five milliseconds and can gate synchronously while language model judges complete asynchronously for reversible actions. Latency budgets are declared per use case in the policy pack
Policy drift	Medium	Policy is configuration, not code, and every change is written to the ledger with its author. Jurisdiction packs isolate regulatory logic so a legal change is a policy edit rather than a release
Calibration on thin data	Medium	Recalibration refuses to run below a minimum labelled sample and never admits a threshold above the lowest-scoring known-incorrect decision. Phase 1 exists specifically to accumulate labels before enforcement begins
Integration resistance	Medium	Interlock operates at the tool-call boundary and requires no access to model internals or agent rewriting. Phase 1 is observe-only, so the first deployment carries no operational risk and produces the evidence needed to justify enforcement

Adversarial adaptation	Medium	Thresholds move with recalibration rather than remaining static. The injection guard is a dedicated classifier rather than a prompt instruction. Fail-closed handling of irreversible actions means a successful evasion still faces a second independent key
Vendor dependency	Low	The check interface is model-agnostic and roles are configuration. An unavailable judge returns a fail-safe risk score rather than a pass, so degradation increases escalation rather than permitting unverified execution

8 Why this team

Team TwoKey delivered a functioning system rather than a demonstration harness. The prototype runs genuine language models from two independent vendors, executes real parallel verification, writes a genuine tamper-evident audit trail and reports measured detection performance against seeded ground truth. Where our Round 1 concept overstated a figure, specifically a twelve millisecond mesh latency that holds only for the deterministic checks, we have corrected it publicly in the README and explained the engineering response rather than restating the original claim.

The judgement we would bring to a client engagement is visible in the artefact. We measured a 41.2 percent false positive rate on our first evaluation, recognised it as the alert fatigue failure mode the brief warns about, diagnosed the cause as an overly strict two-key amount tolerance, and reduced it to 14.3 percent without losing a single true detection. That sequence, measure, diagnose, correct, report, is the working method the proposal depends on.

Financial figures in section 4 are illustrative and modelled on the stated assumptions. Detection, latency and unit cost figures are measured from the prototype evaluation harness and are reproducible with `make eval`. All data used in the prototype is synthetic.